

# Automated Circuit Discovery for NIDS

## Mechanistic Interpretability for SOC-Oriented Security Research

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# Presentation Outline

- 1 Motivation & Background
- 2 Technical Framework
- 3 Research Applications
- 4 SOC-Oriented Integration
- 5 Conclusion

# The Interpretability Gap in NIDS

- Modern NIDS use Deep Learning (Transformers/RNNs) for high accuracy.
- **The Black-Box Problem:** Analysts receive alerts but cannot verify the causal logic.
- **SOC Impact:** High false-positive rates and lack of trust in "AI-driven" security.
- **Solution:** Mechanistic Interpretability to reverse-engineer model internals.
- This motivates causal—not correlational—explanations.

# Interpretability Methods in NIDS: A Comparison

| Aspect                | Feature Attribution<br>(SHAP / LIME) | Rule-Based XAI      | ACDC (Proposed)                   |
|-----------------------|--------------------------------------|---------------------|-----------------------------------|
| Explanation Level     | Input features                       | Human-defined rules | Internal computational components |
| Causality             | Correlational                        | Assumed by rules    | Causal (via activation patching)  |
| Model Internals       | Ignored                              | Not applicable      | Explicitly analyzed               |
| Attack-Specific Logic | Weak                                 | Manual              | Automatically extracted           |
| SOC Usefulness        | Limited                              | High but rigid      | High and operationally actionable |
| Forensic Value        | Low                                  | Moderate            | High (circuit-level audit)        |

# Applying ACDC to NIDS: Core Idea

- In NIDS, the target behavior is an **attack detection decision** (e.g., SYN flood, scan, exfiltration).
- ACDC treats the trained NIDS as a **signal-processing system** that transforms raw traffic statistics into security decisions.
- The objective is to identify the **minimal internal computation** that maps:
  - traffic patterns (rates, flags, temporal bursts)
  - to an attack-specific alert.
- The extracted circuit reveals **which traffic characteristics** and **which internal interactions** the NIDS truly relies on.

# Mapping ACDC Concepts to NIDS

| ACDC Concept        | Interpretation in NIDS                                        |
|---------------------|---------------------------------------------------------------|
| Behavior            | Specific attack type (e.g., DDoS, Scan, Brute Force)          |
| Computational Graph | NIDS model layers and feature interactions                    |
| Activation Patching | Replacing malicious traffic activations with benign baselines |
| Edge Pruning        | Removing causally irrelevant feature pathways                 |
| Circuit             | Minimal computation responsible for attack detection          |
| Faithfulness Metric | KL divergence between full and pruned model outputs           |

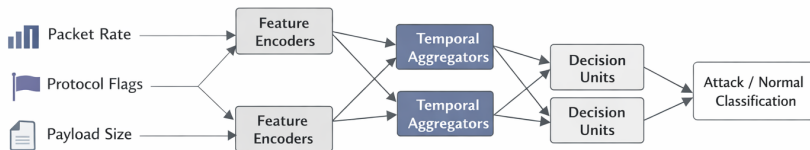
# The NIDS Computational Graph

- The trained NIDS is modeled as a **directed computational graph**.
- **Nodes ( $V$ )**: internal processing units such as feature interaction layers, temporal aggregation modules, attention heads, or hidden neurons.
- **Edges ( $E$ )**: linear information pathways that propagate traffic signals (e.g., packet rates, flags, burst patterns) across the network.
- **Objective**: identify and retain only those edges that are **causally necessary** for a specific attack detection behavior.

This abstraction enables circuit discovery across different NIDS architectures (Transformer, RNN, CNN).

# System Architecture: ACDC-Enabled NIDS

## NIDS as a Computational Graph





# Step A: Task and Dataset Setup

To find a circuit, we must define the behavior:

## Task Definition

Detecting SYN Flood attacks within the UNSW-NB15 dataset.

- **Clean Dataset:** Malicious traffic samples.
- **Corrupted Dataset:** Normal baseline traffic or mean-ablated features.

Behavior is quantified using the logit difference between Attack and Normal classes.

## Step B: Pruning Criterion ( $\tau$ )

ACDC removes an internal connection ( $w \rightarrow v$ ) if it does not causally influence the attack detection outcome.

$$D_{KL}(G \parallel H_{\text{new}}) - D_{KL}(G \parallel H) < \tau$$

- $G$ : Full NIDS model predicting attack vs. normal traffic.
- $H$ : Current candidate circuit under evaluation.
- $D_{KL}$ : Change in output probability distribution.

KL divergence ensures the circuit preserves the **decision logic**, not merely classification accuracy.

## Step C: Activation Patching

- Internal activations from **corrupted (benign) traffic** are injected into the forward pass of **malicious traffic**.
- If the attack prediction remains unchanged, the patched connection is **not causally required** for detection.
- This allows ACDC to distinguish **true attack logic** from incidental correlations.
- **Example:** replacing SYN packet-rate activations with statistics from normal background traffic.

# Research: Generalization vs. Spurious Correlation

- Does the NIDS learn **attack-relevant traffic patterns** (e.g., TCP flags, packet-rate bursts, temporal structure)?
- Or does it rely on **incidental identifiers** (e.g., source IPs, ports, or dataset-specific artifacts)?
- ACDC reveals whether the extracted circuit encodes **robust attack semantics** or shortcut-based correlations.
- This analysis helps distinguish **true generalization** from spurious detection behavior.

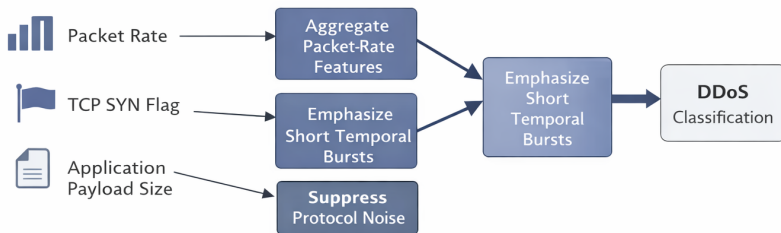
## Example: Circuit Discovery for DDoS Detection

- Consider a trained NIDS tasked with detecting **volumetric DDoS attacks**.
- ACDC isolates a **minimal detection circuit** that:
  - aggregates packet-rate signals across short time windows,
  - emphasizes sudden temporal traffic bursts,
  - suppresses protocol-specific and benign background noise.
- This circuit constitutes the **causal computation** sufficient for triggering a DDoS alert.

# ACDC-Based Circuit Extraction for DDoS Attacks

## Example: Circuit Discovery for DDoS Detection

ACDC isolates a DDoS-specific circuit within the trained NIDS



- Adversarial traffic can induce **pathway divergence**, activating internal circuits normally associated with benign traffic.
- **Research Question:** To what extent can adversarial perturbations reroute computation away from the attack-specific circuit while preserving normal-traffic pathways?
- ACDC enables **circuit-level robustness analysis** by explicitly identifying which internal pathways are disrupted.

# Technical Challenge: Scalability

- Activation patching is computationally expensive for high-throughput network traffic.
- **Research Focus:** Attribution patching to approximate circuit importance with significantly reduced computational cost.
- As a result, ACDC is currently suited for **offline analysis and post-incident investigation**, rather than inline detection.

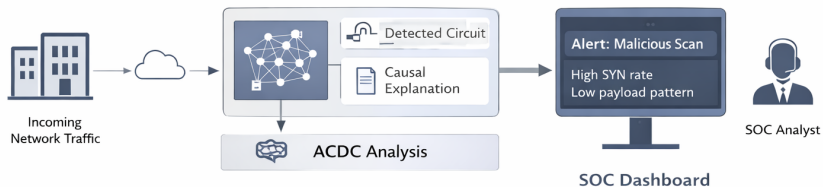


# Failure Case: The OR-Gate Problem

- A NIDS may contain **redundant internal pathways** that independently support the same attack detection.
- If removing one pathway does not degrade detection performance, ACDC may incorrectly prune both during causal testing.
- This does not indicate model failure, but highlights **redundancy in the learned detection logic**.
- Addressing such redundancy remains an open research problem in circuit discovery methods.

# System Design: ACDC-Enabled NIDS

## SOC-oriented Explainable Alerts



# SOC Integration: Transparent Alerts

Instead of: *Alert: Anomaly detected*

## ACDC-Enhanced Alert

Logic: Observed high SYN count combined with low data payload patterns.

Causal confidence: 99.2%.

Circuits are summarized, not shown raw, to analysts.

# Automated Forensics for Tier-3 Analysts

- For each detected attack, ACDC extracts the **internal detection circuit** used by the NIDS.
- Analysts can inspect the **behavioral logic** (e.g., traffic patterns and internal interactions) that triggered the alert.
- This supports **faster root-cause analysis** for advanced and previously unseen threats.

# Logic-Based Filter Tuning and Evaluation

- Circuit inspection identifies causes of false positives.
- Model refinement is applied only to implicated sub-components.
- Evaluation metrics:
  - KL divergence under ablation,
  - detection accuracy preservation,
  - circuit sparsity.

# Building Operational Trust

- **Auditability:** ACDC provides a causal explanation for each intrusion detection decision.
- **Accountability:** Enables post-incident review and justification in regulated and security-critical environments.
- **Inspectability:** Improves analyst confidence by making AI-driven decisions transparent rather than opaque.

- Framed Network Intrusion Detection as a **mechanistic interpretability problem** rather than a black-box prediction task.
- Applied Automatic Circuit Discovery (ACDC) to extract **causal, attack-specific computational circuits** from trained NIDS models.
- Demonstrated how circuit-level explanations enable **auditable, trust-oriented decision support** for SOC operations.

- Scalable and approximate ACDC methods for high-speed network environments.
- Circuit discovery under redundancy and overlapping attack logic.
- Circuit-to-rule translation for deployment in SOC workflows.
- Behavioral analysis of zero-day attacks using circuit-level explanations.
- Human-in-the-loop refinement of interpretability for trusted NIDS.



Thank You!